Tyler Chandross-Cohen

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Summary

Food microbiology researcher with a strong record of conducting interdisciplinary research at the interface of microbial genomics, molecular biology, and food safety. My work centers on the foodborne pathogen *Bacillus cereus*, with a particular focus on how genetic diversity and environmental conditions govern virulence expression and cytotoxic potential across diverse isolates, including psychrotolerant strains relevant to refrigerated foods. By integrating whole-genome sequencing with *in vitro* human cell-based cytotoxicity models, I assess virulence potential of *B. cereus* isolates originating from diverse environmental and food-associated sources. I employ integrated genomic, transcriptional, and phenotypic approaches to characterize *B. cereus* under food-relevant conditions, linking molecular signatures with functional outcomes in human intestinal epithelial cell models. In parallel with my research, I am actively engaged in mentoring undergraduate researchers and teaching.

Education and Training

Institution	***Degree***	***Year***	***Field of Study***	***GPA***	***Relevant Coursework***
The Pennsylvania State University	Ph.D.	TBD	Food Science	3.98	Applied Bioinformatics (852), Applied Statistics (500), Professional Skills for Graduate Students (503), Research Methods (501).
The Pennsylvania State University	B.S.	2022	Food Science with a minor in Microbiology	3.68*	Food Microbiology (408), Food Microbiology Lab (409), Communicating Research in Agricultural Sciences (422), Food Engineering (405), Food Law (497), Medical Microbiology (412), Microbial Genomic Epidemiology (517).

* Dean's List: Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Spring 2022

Research Experience

Experience	***Year***
The Pennsylvania State University Food Science Department	*Aug 2022 – Present*
Graduate Research Assistant	

- Conduct doctoral research on the foodborne pathogen *Bacillus cereus*, with emphasis on psychrotolerant isolates and targeted gene knockout strategies to better understand virulence gene mediated cytotoxicity.
- Design and perform growth experiments in skim milk broth across multiple temperature schemes to inform quantitative exposure assessment models.
- Quantify cytotoxicity of *B. cereus* isolates using human intestinal epithelial cell culture models to evaluate virulence potential under food-relevant conditions.
- Train and mentor undergraduate researchers; oversee laboratory operations including media preparation; and develop detailed laboratory protocols, standard operating procedures (SOPs), and experimental proposals.

The Pennsylvania State University Food Science Department *Jan 2021 – May 2022*
Undergraduate Research Associate

- Collaborated with a graduate student researcher to detect and quantify *Listeria monocytogenes* from environmental samples collected in ice cream manufacturing facilities.
- Isolated and characterized protective cultures and evaluated their inhibitory activity against *L. monocytogenes* using standard microbiological and analytical assays.

The Pennsylvania State University Food Science Department *Oct 2018 – May 2022*
Undergraduate Research Assistant

- Conducted undergraduate research in the laboratory of Dr. Molly Kelly, focusing on experimental evaluation of the impact of the spotted lanternfly on wine quality and safety.
- Designed and performed experiments to assess effects of spotted lanternfly infestation on wine fermentation, chemical composition, and overall product quality.
- Analyzed yeast strain performance with respect to aroma and flavor development through controlled fermentation experiments and sensory-related analytical assays.
- Monitored daily wine production activities, including systematic sampling, chemical testing, and quantitative data collection and analysis.

The Pennsylvania State University Plant Science Department *Sept 2019 – Mar 2020*
Undergraduate Research Assistant

- Assisted with research projects in the laboratory of Dr. Michela Centinari, supporting viticulture and enology-focused experimental studies.
- Conducted experiments evaluating the effects of budburst-delay strategies in grapevines on grape composition and downstream wine chemistry.
- Performed berry sampling, grape juice chemical analyses, and carbohydrate profiling; managed data entry; assisted with grape processing; and monitored wine fermentations.

Teaching Experience

Experience ***Semester***

The Pennsylvania State University Food Science Department *Fall 2025*
Instructor on Record

- Taught two laboratory sections per day, twice weekly (Tuesdays and Thursdays), providing instruction to undergraduate students on microbiological methods in food safety and quality.
- Delivered pre-lab lectures and guided students through exercises involving isolation, enumeration, and identification of foodborne pathogens and spoilage microorganisms.
- Implemented and maintained pre-established course curriculum, protocols, and assessments.
- Supervised and mentored students in aseptic technique, serial dilutions, plate counting, enrichment and selective culturing, and biochemical characterization.
- Managed biosafety and laboratory operations, ensuring compliance with BSL-2 practices and appropriate waste handling.
- Evaluated student progress through lab reports, quizzes, and practical exams.

The Pennsylvania State University Food Science Department *Spring 2023, Spring 2024,*
Short Course Graduate Teaching Assistant

- Teaching Assistant for the Ice Cream Short Course. *Spring 2025, Spring 2026*
- Assisted in laboratory in ice cream freezing, mix calculation and vanilla flavor.
- Assisted in homework help sessions to ensure the success of the students.

The Pennsylvania State University Food Science Department *Fall 2023, Fall 2024*
Graduate Teaching Assistant
- Graduate Teaching Assistant for Food Microbiology Laboratory (FDSC 409).
- Present in class to assist in class discussions and troubleshooting student's experiments.
- Hosted weekly office hours to guide students and answer their questions.
- Graded laboratory reports, results sheets, and exams

The Pennsylvania State University Chemistry Department *Fall 2020, Spring 2021, Fall 2021, and Spring 2022*
Learning Assistant and Grader
- Hosted two weekly LA sessions to review problem sets with students.
- Supported outside-of-classing learning and gave extra help to students.
- Graded midterm and final exams

The Pennsylvania State University Food Science Department *Fall 2021*
Undergraduate Teaching Assistant
- Undergraduate Teaching Assistant for Food Microbiology (FDSC 408).
- Present in class to answer questions and was involved in class discussions.
- Hosted weekly office hours to answer student's questions and support their learning. Reviewed and proctored exams.

Peer Reviewed Publications

Citation	***Contribution***
Chandross-Cohen, T., Ebersole, M., Rahman, N., Wagner, A. S., Kovac, J., & Wildschutte, H. (2026). "Whole-genome Sequence and Phenotypic Characteristics of *Bacillus cereus* Strain IM8 Isolated from the Upper Fremont Glacier in Wyoming," *Microbiology Resource Announcements*. DOI: https://doi.org/10.1128/mra.01260-25	Conducted experiments, Writing original draft, Writing review & editing, Visualization
Nguyen, P., **Chandross-Cohen, T.**, Kovac, J., & Restanio, L. (2025). "Isolation and Characterization of *Bacillus cereus*Group Species in Powdered Infant Formula and Infant Cereal Using a Newly Developed Detection System," *Journal of Food Protection*. DOI: https://doi.org/10.1016/j.jfp.2025.100677	Conducted experiments, Writing original draft, Writing review & editing, Visualization
Navarre, A., Rupert, K., **Chandross-Cohen, T.**, & Kovac, J. (2025). "Low Prevalence and Concentrations of *Campylobacter* Detected on Retail Chicken Breasts," *Journal of Food Protection*. DOI: https://doi.org/10.1016/j.jfp.2025.100635	Whole genome sequencing of isolates, Writing review & editing
Chandross-Cohen, T., Chung, T., Watson, S., Rolon, M. L., & Kovac, J. (2025). "Precision Food Safety: Advances in Omics-based Surveillance for Proactive Detection and Management of Foodborne Pathogens," *Trends in Food Science & Technology*, 105186. DOI: https://doi.org/10.1016/j.tifs.2025.105186	Writing original draft, Writing review & editing, Visualization.
Su, J*., **Chandross-Cohen, T.***, Qian, C., Carroll, L., Kimble, K., Yount, M., Wiedmann, M., & Kovac, J. (2024). "Assessment of the Exposure to Cytotoxic *Bacillus cereus* Group Genotypes Through HTST Milk	Development of methods, conducted experiments, data analysis, and co-wrote and revised the manuscript

Citation	Contribution
Consumption," *Journal of Dairy Science*. DOI: https://doi.org/10.3168/jds.2024-24703 (*Co-first authors*)	
Wei, X., Hassen, A., McWilliams, K., Peitrzen, K., Chung, T., Mendez Acevedo, M., **Chandross-Cohen, T.**, Dudley, E. G., Vipham, J., Mamo, H., Sisay Tessema, T., Zewdu, A., & Kovac, J. (2024). "Genomic Characterization of *Listeria monocytogenes* and *Listeria innocua* Isolated from Milk and Dairy Samples in Ethiopia," *BMC Genomic Data*, 25, 12. DOI: https://doi.org/10.1186/s12863-024-01195-0	Re-confirmed the *Listeria spp*. at The Pennsylvania State University.
Rolon, M. L., **Chandross-Cohen, T.**, Kaylegian, K. E., Roberts, R. F., & Kovac, J. (2023). "Context Matters: Environmental Microbiota of Ice Cream Processing Facilities Affects the Inhibitory Performance of Two Lactic Acid Bacteria Against *Listeria monocytogenes*," *ASM Microbiology Spectrum*, 12(1). DOI: https://doi.org/10.1128/spectrum.01167-23	Conducted experiments, conducted data analysis, and co-wrote and revised the manuscript

Publications Under Review

Citation	***Contribution***
Michel, P., Maxson, T., Chivukula, V., Overholt, W., Medina Cordoba, L. K., Ayodele-Abiola, S., McQuiston, J., Beesley, C. A., Bell, M., Caban Figueroa, V., Bugrysheva, J., **Chandross-Cohen, T.**, Weiner, Z., Carroll, L., Kovac, J., & Sue, D. (2026). "Antimicrobial Resistance Profiling and Phenotypic Characterization of Archived Clinical *Bacillus paranthracis* Strains," *ASM Microbiology Spectrum*, in review.	Conducted experiments, Writing review & editing,
Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Praul, B., Weidner, L., & Kovac, J. (2026). "Environmental Modulation of Cytotoxicity and Virulence Factor Stability in Psychrotolerant *Bacillus cereus* Group Isolates," *Food Microbiology*, in review.	Writing original draft, Writing review & editing, Investigation, Formal analysis, Visualization, Data curation, Conceptualization

Poster Presentations

Presentation	***Contribution***
Cavalcante Y., **Chandross-Cohen, T**., Mouratidis, I., Patsakis, M., Georgakopoulos-Soares, I., & Kovac, J. (May 14, 2026). "Culture-independent detection of *Staphylococcus aureus* using multiple displacement amplification and CRISPR-Cas9 coupled with Nanopore sequencing", One Health Microbiome Symposium, Penn State One Health Microbiome Center, University Park, PA	Development of methods, conducted experiments.
Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (March 27, 2026). "Characterization of *Bacillus cereus* Group Isolates from Infant Formula and Cereal by Whole-Genome Sequencing and Cytotoxicity," 2026 Graduate Exhibition, University Park, PA.	Development of methods, conducted experiments, conducted data analysis and **presented**.

Cavalcante, Y., **Chandross-Cohen, T.**, & Kovac, J. (March 27, 2026). "Culture-independent Detection of *Staphylococcus aureus* Using Multiple Displacement Amplification and CRISPR-Cas9 Coupled with Nanopore Sequencing," 2026 Graduate Exhibition, University Park, PA.
Development of methods, conducted experiments, conducted data analysis

Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (March 26, 2026). "Characterization of *Bacillus cereus* Group Isolates from Infant Formula and Cereal by Whole-Genome Sequencing and Cytotoxicity," Gamma Sigma Delta Symposium, University Park, PA. ***Poster Award***.
Development of methods, conducted experiments, conducted data analysis and **presented**.

Cavalcante, Y., **Chandross-Cohen, T.**, & Kovac, J. (March 26, 2026). "Culture-independent Detection of *Staphylococcus aureus* Using Multiple Displacement Amplification and CRISPR-Cas9 Coupled with Nanopore Sequencing," Gamma Sigma Delta Symposium, University Park, PA.
Development of methods, conducted experiments, conducted data analysis

Weidner, L., **Chandross-Cohen, T.**, Fratnik Steyer, A., Petrovic, Z., & Kovac, J. (March 26, 2026). "Genotypic and Phenotypic Characterization of *Bacillus cereus* Isolates from Human Clinical Cases in Slovenia," Gamma Sigma Delta Symposium, University Park, PA. ***Poster Award***.
Development of methods, conducted experiments, conducted data analysis

Lyons, M., Biernbaum, E., **Chandross-Cohen, T.**, Cavalcante, Y., Kovac, J., & Dudley, E. (November 7–8, 2025). "Targeted Nanopore Sequencing of the O-Antigen Region in *E. coli* Using CRISPR Cleavage at gndA," Allegheny Branch of the American Society of Microbiology Annual Meeting, California, PA.
Development of methods, conducted experiments, conducted data analysis

Chandross-Cohen, T., Praul, B., & Kovac, J. (May 6–8, 2025). "Effects of Protein Stress Factors on the Cytotoxicity of Supernatants from Psychrotolerant *Bacillus cereus* Isolates," IAFP European Symposium, Madrid, Spain.
Development of methods, conducted experiments, conducted data analysis and **presented**.

Chandross-Cohen, T., Praul, B., & Kovac, J. (March 27, 2025). "Effects of Protein Stress Factors on the Cytotoxicity of Supernatants from Psychrotolerant *Bacillus cereus* Isolates," Gamma Sigma Delta Symposium, University Park, PA. Poster Award.
Development of methods, conducted experiments, conducted data analysis and **presented**.

Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Centeno, C., & Kovac, J. (November 1–2, 2024). "Cytotoxicity Assessment of Psychrotolerant *Bacillus cereus* Isolates Across Varied Temperatures," Allegheny Branch of the American Society of Microbiology Annual Meeting, Greensburg, PA.
Development of methods, conducted experiments, conducted data analysis, and **presented**.

Praul, B., **Chandross-Cohen, T.**, & Kovac, J. (November 1–2, 2024). "Effects of Protein Stress Factors on the Cytotoxicity of Psychrotolerant *Bacillus cereus* Isolates," Allegheny Branch of the American Society of Microbiology Annual Meeting, Greensburg, PA.
Development of methods, conducted experiments, conducted data analysis.

Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Centeno, C., & Kovac, J. (July 15–17, 2024). "Cytotoxicity Assessment of Psychrotolerant *Bacillus cereus* Isolates Across Varied Temperatures," International Association for Food Protection Annual Meeting, Long Beach, CA.
Development of methods, conducted experiments, conducted data analysis, and **presented**.

Chandross-Cohen, T., & Kovac, J. (March 27, 2024). "Variation in *Bacillus cereus* Hemolysin BL Transcription Associated with a SNP in the Transcription Regulatory Region," Gamma Sigma Delta Symposium, University Park, PA.
Development of methods, conducted experiments, conducted data analysis, and **presented**.

Yount, M., **Chandross-Cohen, T.**, Georgiana, G., Readinger, E., & Kovac, J. (March 27, 2024). "Surface Colonization Area Was Associated with Phylogenetic Group and Hemolysin BL Production in *Bacillus cereus* sensu lato," Gamma Sigma Delta Symposium, University Park, PA. ***Poster Award***.	Development of methods, assistance with experiments, conducted data analysis.
Chandross-Cohen, T., & Kovac, J. (March 20, 2024). "Variation in *Bacillus cereus* Hemolysin BL Transcription Associated with a SNP in the Transcription Regulatory Region," Food Science in Action Symposium, University Park, PA.	Development of methods, conducted experiments, conducted data analysis, and **presented**.
Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (November 3–4, 2023). "Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model," Allegheny Branch of the American Society of Microbiology Annual Meeting, Williamsport, PA.	Development of methods, conducted experiments, conducted data analysis, and **presented**.
Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (July 16–19, 2023). "Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model," International Association for Food Protection Annual Meeting, Toronto, Ontario, Canada.	Development of methods, conducted experiments, conducted data analysis, and **presented**.
Su, J., Qian, C., **Chandross-Cohen, T.**, Yount, M., Wiedmann, M., & Kovac, J. (July 16–19, 2023). "An Exposure Assessment of Cytotoxic *Bacillus cereus* Strains from Various Phylogenetic Groups in HTST Milk," International Association for Food Protection Annual Meeting, Toronto, Ontario, Canada.	Provided data used in growth modeling.
Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (May 26, 2023). "Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model," Food Science in Action Symposium, University Park, PA.	Development of methods, conducted experiments, conducted data analysis, and co-**presented**.
Chandross-Cohen, T., Yount, M., Su, J., Qian, C., Wiedmann, M., & Kovac, J. (March 30, 2023). "Growth Potential of *Bacillus cereus* Group Strains from Different Phylogenetic Groups in a Dairy Food Model," Gamma Sigma Delta Symposium, University Park, PA. ***Poster Award***.	Development of methods, conducted experiments, conducted data analysis, and **presented**.
Rolon, M. L., **Chandross-Cohen, T.**, Kaylegian, K. E., Roberts, R. F., & Kovac, J. (November 4–5, 2022). "Context Matters: Environmental Microbiota of Ice Cream Processing Facilities Affects the Inhibitory Performance of Two Lactic Acid Bacteria Against *Listeria monocytogenes*," Allegheny Branch of the American Society of Microbiology Annual Meeting, Morgantown, WV.	Conducted experiments, conducted data analysis, and **presented**
Rolon, M. L., **Chandross-Cohen, T.**, Kaylegian, K. E., Roberts, R. F., & Kovac, J. (July 31–August 3, 2022). "Context Matters: Environmental Microbiota of Ice Cream Processing Facilities Affects the Inhibitory Performance of Two Lactic Acid Bacteria Against *Listeria monocytogenes*," International Association for Food Protection Annual Meeting, Pittsburgh, PA.	Conducted experiments and conducted data analysis
Chandross-Cohen, T., Rolon, M. L., & Kovac, J. (March 31, 2022). "Isolation, Characterization, and Application of Antilisterial Isolates in a Raw Milk Cheese Model to Inhibit *Listeria monocytogenes*," Gamma Sigma Delta Symposium, University Park, PA. ***Poster Award***.	Original ideas, conducted experiments, conducted data analysis, and **presented**.

Oral Presentations

Presentation	*Contribution*
Chandross-Cohen, T., Nguyen, P. T., Restaino, L., & Kovac, J. (May 14–15, 2026). "Cereus-ly a Concern: Genomic and Cytotoxic Characterization of *Bacillus cereus* Group Isolates from Infant Foods," Pennsylvania Association for Food Protection Annual Meeting, University Park, PA. ***Invited***	Development of methods, conducted experiments, conducted data analysis and **presented**.
Chandross-Cohen, T., Yount, M., Readinger, E., Prince, C., Kimble, K., Praul, B., Centeno, C., & Kovac, J. (July 28–30, 2025). "Cytotoxicity, Transcription, and Stability of Virulence Factors Produced by Psychrotolerant *Bacillus cereus* Group Isolates," International Association for Food Protection Annual Meeting, Cleveland, OH**.**	Development of methods, conducted experiments, conducted data analysis, and **presented**.

Mentorship Experience

Student	*Mentorship Period*
Lydia Weidner *	*Nov 2024 – Current*
Brian Praul	*July 2024 – August 2025*
Kailee Shotto *	*Aug 2023 – May 2024*
Elizabeth Henry *	*Aug 2023 – Jan 2024*
Erin Readinger *	*Jan 2023 – May 2024*
Mackenna Yount *	*Aug 2022 – May 2024*
Express Mentorship Group at Rutgers Preparatory School	*Aug 2021 – Current*

*Undergraduate Student at Penn State

Leadership Experience

Role	*Date*
President of the Penn State Chapter of Phi Tau Sigma	*Apr 2023 – Current*
Kovac Lab Safety Coordinator	*Jan 2023 – Current*
Treasurer of the IAFP Student Professional Development Group	*May 2023 – July 2025*
Treasurer of the Penn State Chapter of Phi Tau Sigma	*May 2022 – Apr 2023*

Awards and Honors

Award/Honor	*Year*	*Funding Source*
Gamma Sigma Delta Symposium First Place Winner Food Systems Graduate Division	*2026*	*
Gamma Sigma Delta Symposium First Place Winner Microbiology/Microbiome Graduate Division	*2025*	*
Dr. Daryl B. & Mrs. Dawn L. Lund Student International Travel Scholarship of Phi Tau Sigma	*2025*	◊
Recipient of the WH Pierce Prize for Global Impact in Microbiology, One Health Microbiome Center, Presented by Applied Microbiology International.	*2024*	*
John H Hetrick Food Science Scholarship	*2024*	*
International Association for Food Protection Student Travel Award	*2024*	◊

Allegheny Branch, American Society of Microbiology Meeting Second Place Winner Graduate Division Poster presentation	*2023*	◊
Dudek Graduate Scholarship in Food Science	*2023*	*
Internal Agricultural Sciences Scholarship	*2023*	*
Food Industry Group Graduate Student Leadership Award	*2023*	*
Gamma Sigma Delta Symposium First Place Winner Graduate Division	*2023*	*
Internal Agricultural Sciences Scholarship	*2023*	*
PNC Financial Corporation Graduate Fellowship	*2023*	*
Robert Graham Endowment Graduate Fellowship	*2022*	*
Keeney Food Science Department Head Fund	*2022*	*
University Graduate Fellowship	*2022*	*
E&W Rosskam Scholarship in Food Science	*2022*	*
Henry Pierce in Ag Science Scholarship	*2022*	*
Gamma Sigma Delta Symposium First Place Winner Undergraduate Division	*2022*	*
Virginia Dare Award	*2021*	◊
Zepp Family Food Safety Endowment	*2021*	*
Henry Pierce in Ag Science Scholarship	*2021*	*
American Society for Enology and Viticulture National Scholarship	*2021*	◊
Featured Author in Penn Statements Volume 39, Spring 2020 Edition	*2020*	*
Zepp Family Food Safety Endowment	*2020*	*
Keystone Institute of Food Technologists Scholarship	*2020*	◊
Henry Pierce in Ag Science Scholarship	*2020*	*
American Society for Enology and Viticulture National Scholarship	*2020*	◊
National Association of Flavors & Food Ingredient Systems Scholarship Award	*2019*	◊
Featured Student in the Penn State College of Ag Sciences "Ag Student Journeys"	*2019*	*

*Penn State Affiliated Scholarship/Fellowship/Award, ◊ External Scholarship/Fellowship/Award

Competitive Funding

Date	*Agency*	*Lead PI*	*Title*	*Role*	*Amount*	*Credit*	*Funded*
2024-2027	USDA NIFA Predoctoral Fellowship	Tyler Chandross-Cohen	#Assessing The Role of Hemolysin BL in Cytotoxicity and Fitness of *B. cereus* in Food Systems	PI	$180,000	100%	Yes
2023-2024	The Pennsylvania State University College of Agricultural Sciences	Tyler Chandross-Cohen	#Assessing the Effect of Polymorphisms in *hbl* Genes in *B. cereus* on Transcription of *hbl* and Cytotoxicity	PI	$3,000	100%	Yes
2023-2026	NSF Graduate Research Fellowship Program (GRFP)	Tyler Chandross-Cohen	Characterizing the Ecological Role of *B. cereus* Hemolysin BL in biofilm formation and competition	PI	$160,000	100%	No

2021-2022	The Pennsylvania State University Department of Food Science	Tyler Chandross-Cohen	*Evaluating the Antilisterial Activity of Protective Cultures in Food Processing Environmental *in vitro* Biofilms	PI	$2,000	100%	No
2021	The Pennsylvania State University Department of Food Science	Tyler Chandross-Cohen	*Isolation, Characterization, and the Use of Protective Cultures in a Raw Milk Cheese Model to Inhibit *Listeria monocytogenes*	PI	$6,000	100%	Yes

*Undergraduate Research Funding, #Graduate Research Funding

Certifications

Title	***Year***
CITI OSHA Bloodborne Pathogens	*2023*
Hazard analysis and critical control points (HACCP)	*2022*
CITI Biosafety and Biosecurity (BSS)	*2019*
Responsible Alcohol Management Program (RAMP)	*2018*
Laboratory and Research Training	*2018*

Memberships

Organization	***Year***
International Association for Food Protection (IAFP)	*2022 – Current*
Penn State One Health Microbiome Center	*2022 – Current*
American Society for Microbiology (ASM)	*2022 – Current*
Allegany Branch of the American Society for Microbiology (ABASM)	*2022 – Current*
Penn State Chapter of Phi Tau Sigma	*2021 – Current*
Phi Tau Sigma National Honors Society	*2021 – Current*
Institute of Food Technologists (IFT)	*2018 – Current*
Penn State Food Science Club	*2018 – Current*